

Epistemic Decision Theory with Imprecise Credences

We are interested in cases of epistemic *entanglement*: cases in which how the world is depends partly on how an agent thinks it is. Our central example:

Leap: Bob stands on the brink of a chasm, summoning up the courage to try and leap across it. Confidence helps him in such situations: specifically, for any value of x between 0 and 1, if Bob attempted to leap across the chasm while having degree of belief x that he would succeed, his chance of success would then be x . What credence in success is it epistemically rational for Bob to have?

Two competing intuitions in *Leap*:

1. *Intuition One*: Only credence 0 and 1 are epistemically rational for Bob to have, since only at these points is he maximally accurate.
2. *Intuition Two*: Any credence is epistemically rational for Bob to have, since his credences match his chances no matter what.

Competing versions of “Epistemic Decision Theory” predict one or another of these intuitions. We will argue that neither is correct. We think the only state it is rational for Bob to have is an *imprecise* credal state, which includes every value in the range $[0, 1]$.

1. Leaping into Epistemic Decision Theory

Practical Decision Theory models agents with a credence function cr and a value function V and says that rational agents deciding between acts $\phi_1, \phi_2, \dots$ choose the act that maximizes *expected value*.

Standard Practical Decision Theory: if $cr(w \parallel \phi)$ is the agent’s credence that w would be the actual world if they chose ϕ and $V(\phi, w)$ is the value of choosing ϕ if w is the actual world, rational agents choose a ϕ that maximizes

$$\sum_{w \in W} cr(w \parallel \phi) \cdot V(w, \phi). \quad (\text{MAX})$$

Epistemic Decision Theory¹ uses the tools of Practical Decision Theory to evaluate *epistemic* rationality. We replace the value function with an accuracy scoring rule A , which takes a credence function and a world as input and outputs an “accuracy score” that measures how “close to the truth” the credence function is.² Then we say that rational agents deciding between credal states $\pi_1, \pi_2, \dots$ choose the credal state that maximizes *expected accuracy*.

Greaves’ Standard Epistemic Decision Theory (E-MAX): if $cr(w \parallel \phi[\pi])$ is the agent’s credence that w would be the actual world if they had credence function π and $A(\pi, w)$ is the accuracy of π if w is the actual world, epistemically rational agents adopt a π that maximizes

$$\sum_{w \in W} cr(w \parallel \phi[\pi]) \cdot A(w, \pi). \quad (\text{E-MAX})$$

E-MAX says that only credences 1 and 0 are rational for Bob in *Leap*.³

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Greaves (2013), “Epistemic Decision Theory”. We focus on *Leap*, but we think our solution has the best verdicts for Greaves’ other cases, like *Promotion* and *Imps*. Here is *Promotion*, in case it becomes relevant:

Alice is up for promotion. Her boss, however, is a deeply insecure type: he is more likely to promote Alice if she comes across as lacking in confidence. Furthermore, Alice is useless at play-acting, so she will come across that way iff she really does have a low degree of belief that she’s going to get the promotion. Specifically, the chance 1 of her getting the promotion will be $1 - x$, where x is whatever degree of belief she chooses to have in the proposition P that she will be promoted. What credence in P is it epistemically rational for Alice to have?

Here and below when we port over to epistemic decision theory we are using *causal* practical decision theory for simplicity – the differences between causal and evidential decision theory are orthogonal.

¹ This approach is sometimes called an *accuracy-first* epistemology. It has been championed by Joyce (1998), Greaves and Wallace (2006), Leitgeb and Pettigrew (2010), Pettigrew (2016), and others.

² It is standard to take the squared-loss (or ‘quadratic’) scoring rule:

$$A_Q(cr, w) = - \sum_{q \in W} (cr(q) - \mathbf{1}_w(q))^2.$$

³ Subject to some natural choices for the accuracy scoring rule A . Following standard EpDT, we assume that A is *strictly proper*, meaning that every probability function evaluates itself as most accurate in expectation (Gibbard 2008; Joyce 2009).

2. Konek and Levinstein's Act-State Distinction

Konek and Levinstein (2019) (henceforth K&L) argue that $E\text{-MAX}$ does not give a properly *epistemic* rationality judgment, because it fails to distinguish between epistemic states and the acts of adopting them.

- An epistemic **state** is a cognitive state held by an agent, and is epistemically rational just in case it *reflects* the world well.
- An epistemic **act** is the act of adopting an epistemic state, and is practically rational for an agent (who cares only about accuracy) just in case it *causes* the agent to have an accurate epistemic state.

K&L argue that $E\text{-MAX}$ goes wrong because it evaluates rationality by looking at how much accuracy the *act* of adopting an epistemic state brings about, when it should care only about the accuracy of the state itself.

They thus propose an Epistemic Decision Theory based on the *unconditional* decision theory of Leonard Savage wherein credal states are rational just in case, *supposing that they are adopted*, they maximize expected accuracy.

Konek and Levinstein's SAV-MAX : epistemically rational agents adopt a π such that for any other credal state π^* ,

$$\sum_w \pi(w) \cdot A(\pi, w) \geq \sum_w \pi(w) \cdot A(\pi^*, w). \quad (\text{SAV-MAX})$$

Given an additional chance-deference assumption, the resultant theory says that any credence function cr_x is rational for $x \in [0, 1]$.

Konek and Levinstein's CHANCE-MAX : if $ch(w \parallel \phi[\pi])$ is the objective chance that w would be the actual world if the agent had credence function π , epistemically rational agents adopt a π such that for any other credal state π^* ,

$$\sum_w ch(w \mid \phi[\pi]) \cdot A(\pi, w) \geq \sum_w ch(w \mid \phi[\pi]) \cdot A(\pi^*, w). \quad (\text{CHANCE-MAX})$$

CHANCE-MAX says that any credence $x \in [0, 1]$ in success is rational for Bob in *Leap*. It aligns with *Intuition Two*, where $E\text{-MAX}$ aligns with *Intuition One*.

3. CHANCE-MAX does not respect the Act-State Distinction

CHANCE-MAX is not the only way to vindicate *Intuition Two*. But we needn't worry about these variations, because there is a deeper problem:

Problem: CHANCE-MAX does not respect the act-state distinction.

Given the setup of the case, any epistemic act Bob chooses *makes it the case* that his chances of success take on a particular value. Because Bob's chances of success partly determine how accurate he is, CHANCE-MAX cannot help but end up giving Bob positive or negative accuracy points for bringing about more or less accuracy via changing the chances.⁴

The idea is that Epistemic Decision Theory ought not take into account the causal effects of adopting credal states, and Savage's theory accordingly assumes that possibilities are fine-grained enough to already take into account causal effects, so that acts do not have causal effects within the framework.

A notable alternative is in Joyce (2019).

⁴ This objection also applies to Joyce's (2019) theory and an alternative that we defend in other work.

What goes wrong? We agree with K&L that epistemic rationality should be about *states* reflecting the world accurately rather than *actions* bringing about accuracy, and thus that Epistemic Decision Theory should avoid taking the causal effects of adopting credences into account.

But in order to avoid taking causal effects into account, we need to identify an *unconditional* prior for the decision theory.⁵ K&L say that when we *suppose* that π is adopted, we then treat π as the unconditional prior. This simply doesn't work out formally, since it means that *every* probability function satisfies SAV-MAX, so it gives no particular recommendation in *Leap* (or any other case).⁶ K&L thus amend SAV-MAX into CHANCE-MAX, thereby vindicating *Intuition Two*, but end up with a theory that doesn't respect the act-state distinction because it still hasn't properly identified an unconditional prior.

4. The Imprecise Credences Solution

We propose that the *right* way to implement Savage-style Epistemic Decision theory requires finding a well-defined unconditional prior that respects Bob's evidence. And we think that in cases like *Leap*, the prior is *imprecise*.

The *imprecise-credences framework*⁷ argues that sometimes evidence does not narrow down a precise rational judgment in some propositions. In these cases, the framework says, we should model an agent's credal state not with a single credence function, but with a *set* of credence functions, called a **representor**.

We think Bob's prior is just such a case of underdetermined evidence: after all, Bob's evidence leaves open what credence he will have in success, and so it does not warrant any precise level of confidence in what credence he will have. But if his chances of success depend on his confidence, then his evidence also does not warrant any precise level of confidence in his chances of success.

So if we want to find Bob a prior that respects the evidence, this prior should not be precise. Instead, we think Bob's rational prior credal state should be modeled by a representor $\mathcal{R}$ where all of the credence functions in the representor agree about the conditional facts, but have different unconditional probabilities that Bob succeeds:

1. For every $x \in [0, 1]$, every credence function cr in $\mathcal{R}$ is certain that if Bob were to adopt credence $\pi(\text{succceed}) = x$, he would have chance x at success.
2. For every $y \in [0, 1]$, there is some credence function cr_y in $\mathcal{R}$ that thinks Bob has y chance at success.

In other words: we can say that the rational prior credal state for the Bob is the imprecise credal state containing all and only the credence functions that are compatible with the facts as described in *Leap*.⁸

⁵ Otherwise, the only estimates we have are ones that take causal effects into account. Causal Decision Theory uses causal-conditional estimates, but Savage decision theory requires unconditional estimates.

⁶ In fact, in K&L's framework SAV-MAX ends up being a restatement of strict propriety! This is partly why we didn't write out the statement of strict propriety in a footnote.

⁷ Levi (1985), Joyce (2010), Schoenfield (2012) among others

Technically we model our agents with sets of pairs of conditional and unconditional credence functions. In general, suppose an agent S has causal conditional credences conditional on acts they might adopt, modeled by a set K^S of conditional probability functions $x^S(\cdot \parallel \cdot)$, and unconditional credence functions over states of the world, modeled by a set Cr^S of probability functions cr^S . Then we define their representor as $\mathcal{R}^S = Cr^S \times K^S = \{ \langle cr, x \rangle \mid cr \in Cr^S, x \in K^S \}$.

⁸ Greaves (2013) worries about an interpretational issue around identifying a prior credence function in cases like *Leap*. She ends up saying that the prior credence function is picked out as "respecting the specification of the case", but that "there is no time at which the 'initial' credence function is actually possessed." We say that Bob *does* possess a prior: it is the imprecise state containing all the credence functions respecting the specification of the case.

5. The Scope of Epistemic Decision Theory

We introduced imprecise credences in order to find Bob a well-defined prior from which we could properly implement a Savage-style epistemic decision theory. But if rational agents are sometimes representable with imprecise credences, and if our account of Bob's rational prior is correct, then it seems we have already found the epistemically rational credal state for Bob to have. What do we still need Epistemic Decision Theory for?

We think that the question of what epistemic state is rational, given Bob's evidence, is indeed answered *prior* to applying Epistemic Decision Theory. An Epistemic Decision Theory that respects K&L's act-state distinction must start with a well-defined prior, but once we have the well-defined (imprecise) prior, there may be no more work for Epistemic Decision Theory to do.

To be clear: we mean no work for Epistemic Decision Theory to do in cases like *Leap*, with act-state entanglement. The program has a marvelous track record in justifying probabilism and updating norms, and nothing in our thesis undermines that. And Kate brings up some good worries related to this point that we will discuss in a response!

6. Practical Rationality for Epistemic Acts

We can still ask which precise credence function an agent should adopt, but this is to ask what (epistemic) act to take, rather than whether the *resultant state* is epistemically rational. And we think we ought to evaluate acts, epistemic or not, using practical decision theory. We should be asking if these acts are practically rational, not if they are epistemically rational.

So we don't think Bob is *epistemically justified* in adopting any precise posterior, if we take K&L's distinction seriously. In adopting any precise posterior, Bob *brings about* the evidence to support it, and epistemic rationality should not value the additional accuracy brought about. Nonetheless, Bob can be *practically rational* in adopting a precise posterior.

What posterior does practical decision theory say Bob should adopt? This depends on Bob's values! Assuming Bob values making the leap, it will say that the uniquely practically rational option is for Bob to adopt π_1 . (We calculate expected practical value using the conditional portion of Bob's representor, which agree on $cr(\text{succeed} \mid \phi[\pi_x])$ for all π_x .) Assuming Bob values having precise credences aligned with the chances, any precise credence is rational.

So to the question "when is it epistemically rational to adopt a new posterior without new evidence?", we say: not in general, and certainly not in the cases we've considered. But to the question "when is it practically rational to adopt a new posterior without new evidence?", we say: when it maximizes expected (practical) value. Here we follow Joyce (2010, p.292), who suggests that, even if one's evidence only warrants a maximally imprecise state, practical reasons may warrant going beyond one's evidence. We may sharpen in "epistemically defective" ways in order to reap practical benefits (like being able to make decisions). Thus a sharpening strategy "may play a role in practical decision making even if it is defective as epistemology".